Project 9: Automated Warehouse Robot Controller

Aim

Design a multi-agent control system for a fleet of warehouse robots (like Kiva/Amazon Robotics). The system must coordinate multiple robots to move shelves to packing stations without collisions or deadlocks, using Multi-Agent Pathfinding (MAPF) and Cooperative Search techniques.

a. Data Collection & Research

Dataset Overview:

- **Bonus (Local Data):** Visit a local logistics center or large warehouse (e.g., Uno Hypermarket storage, Yalidine, or a university library archive). Sketch their floor plan to create a custom grid map.
  - **Layout:** Create a grid file representing obstacles (shelves) and open space.
  - **Task List:** Generate a "Pick List" based on real observations (e.g., "Items from Aisle 3 are ordered most frequently").
- **Standard (International):** Use the MovingAI Benchmarks, specifically the warehouse maps.

External Resources:

- Research Multi-Agent Pathfinding (MAPF) and the concept of "Reservation Tables" (Time-Space A*).
- Study "Lifelong MAPF" where tasks appear continuously.

b. Problem Definition

Given N robots at start positions and N goal positions (shelves to retrieve), find a set of conflict-free paths such that all robots reach their goals in minimal time.

c. Constraints & Objective Function

Constraints:

- **Hard Constraints:**
  - No two robots can occupy the same cell at the same time t.
  - No "Edge Conflicts": Robot A moves $u \to v$ while Robot B moves $v \to u$.
- **Soft Constraints:**
  - Maintain a safety distance.
  - Minimize waiting (idling).

Objective Function:

- Minimize Makespan: The time until the last robot finishes its task.
- Minimize Flowtime: The sum of time taken by all robots.

d. Search Strategy Implementation

Cooperative A* (Hierarchical):

- Plan paths sequentially.
- Robot 1 plans first (ignoring others).
- Robot 2 plans next, treating Robot 1's path as moving obstacles (dynamic barriers).

Independent A* (Baseline):

- Plan all robots independently.
- If a collision occurs, simple "wait and retry" (expected to fail or deadlock in crowded maps).

Hill Climbing:

- Optimization phase: Take a valid solution and try to shorten paths by swapping move orders.

Conflict-Based Search (CBS) (Bonus):

- Two-level search:
  - High level: Searches the Conflict Tree (resolving collisions).
  - Low level: Finds paths for individual agents.

e. Comparative Evaluation

Performance Comparison:

- Compare the success rate of Independent A* (frequency of deadlock i.e. How often it deadlocks) vs. Cooperative A*.
- Measure the total number of steps for varying fleet sizes (5 robots vs. 20 robots).

Success Criteria:

- Visual demonstration of 10+ robots moving simultaneously without crashing.
- The system handles a head-on collision scenario gracefully (e.g., one robot moves aside).

f. Deliverables

- **Working Prototype:** A grid-based simulation (GUI) showing robots as moving colored dots carrying shelves.
- **Visualizations:** A "Heatmap" of congestion (which corridors are most blocked?).
- **Documentation:** Detailed explanation of the collision-checking logic in the time dimension.